

Hotel Booking Cancellation Prediction Using Machine Learning

Final Machine Learning Assessment Report

Live Application

Streamlit: <https://hotel-cancellation-app.streamlit.app/>

GitHub: https://github.com/abhishekmaher6699/ds_assignment

1. Abstract

This project develops a machine learning system to predict whether a hotel booking will be cancelled. The Hotel Booking Demand dataset was cleaned, explored, transformed through feature engineering, and used to train Logistic Regression, Decision Tree, Random Forest, and Support Vector Machine models. Random Forest was additionally optimized using GridSearchCV with 5-fold cross-validation. On the reported test set, the tuned Random Forest achieved 0.8401 accuracy, 0.7708 precision, 0.5959 recall, and a 0.6722 F1 score.

2. Problem Statement

Hotel cancellations can affect room availability, revenue planning, and operational decisions. The objective is to predict the binary target **is_cancelled**, where 0 represents a booking that was not cancelled and 1 represents a cancelled booking.

3. Dataset and Preprocessing

The workflow includes missing-value and duplicate inspection, data cleaning, exploratory analysis, feature engineering, and preprocessing of numerical and categorical variables. Numerical variables were imputed and scaled, while categorical variables were imputed and one-hot encoded. `reservation_status`, `reservation_status_date`, `agent`, and `company` were excluded from the predictive feature set.

4. Feature Engineering

Three derived features were created: **total_guests** from adults, children, and babies; **total_nights** from weekend and weekday nights; and **total_stay_cost** from ADR multiplied by total nights.

5. Exploratory Data Analysis

The correlation matrix was used to inspect relationships among numerical variables during exploratory analysis.

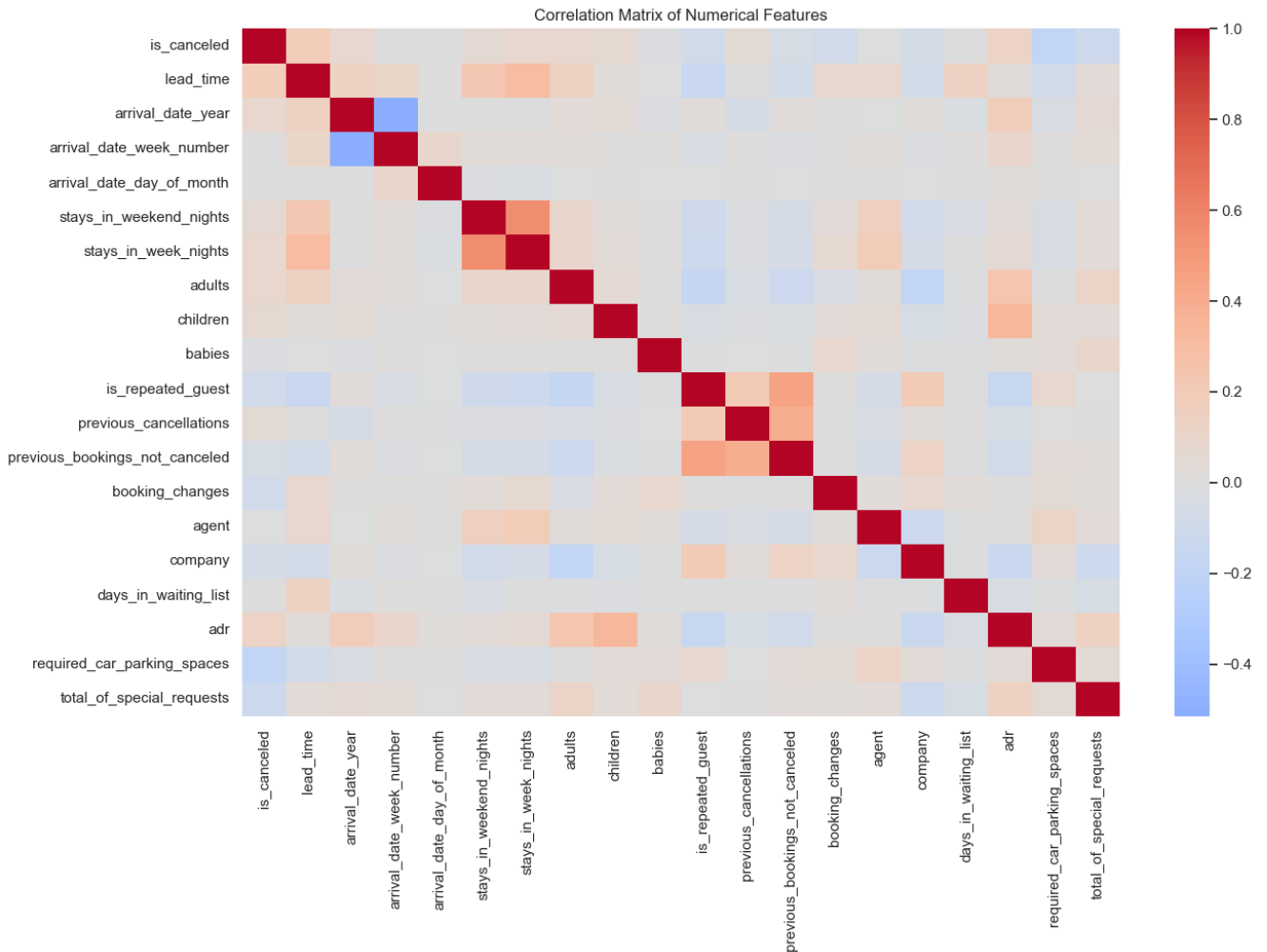


Figure 1. Correlation matrix of numerical features.

6. Model Evaluation

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	0.7955	0.6806	0.4831	0.5651
Decision Tree	0.7929	0.6223	0.6283	0.6253
Random Forest	0.8393	0.7728	0.5890	0.6685
SVM	0.7937	0.6868	0.4590	0.5503
Tuned Random Forest	0.8401	0.7708	0.5959	0.6722

The tuned Random Forest increased test accuracy from 0.8393 to 0.8401 and F1 score from 0.6685 to 0.6722 relative to the original Random Forest.

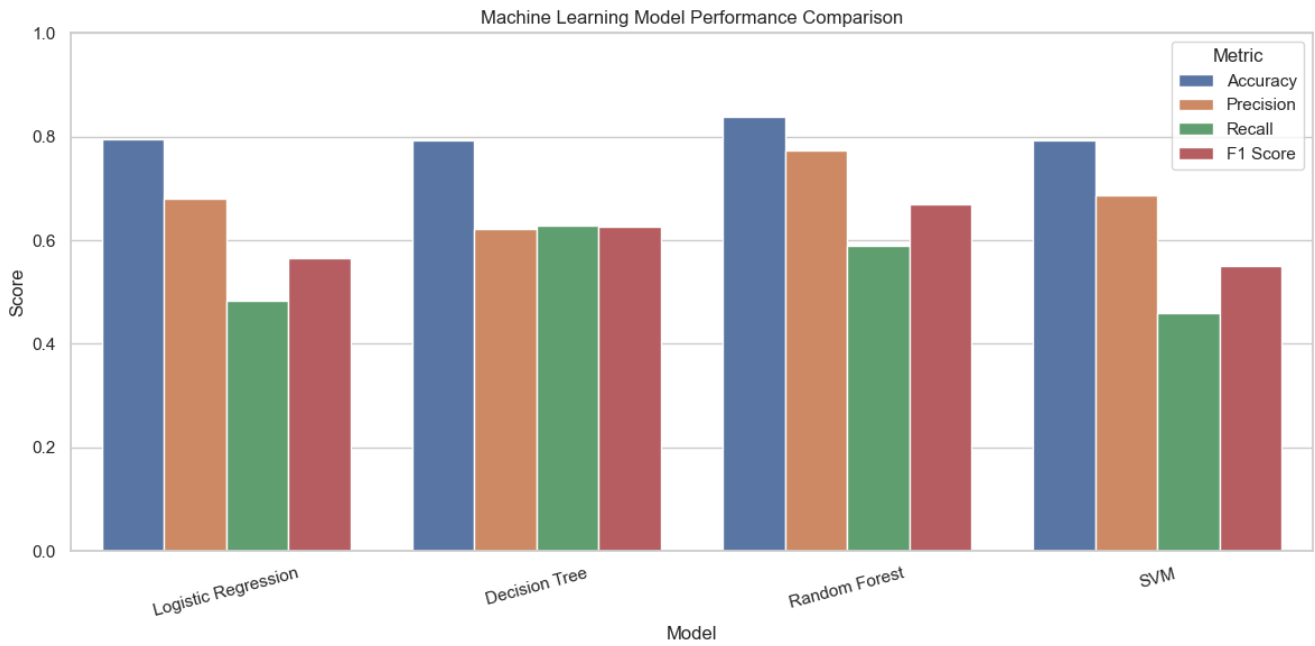


Figure 2. Baseline model performance comparison.

7. Confusion Matrix Analysis

The confusion matrices show correct and incorrect predictions for the four baseline models and provide additional information beyond aggregate accuracy.

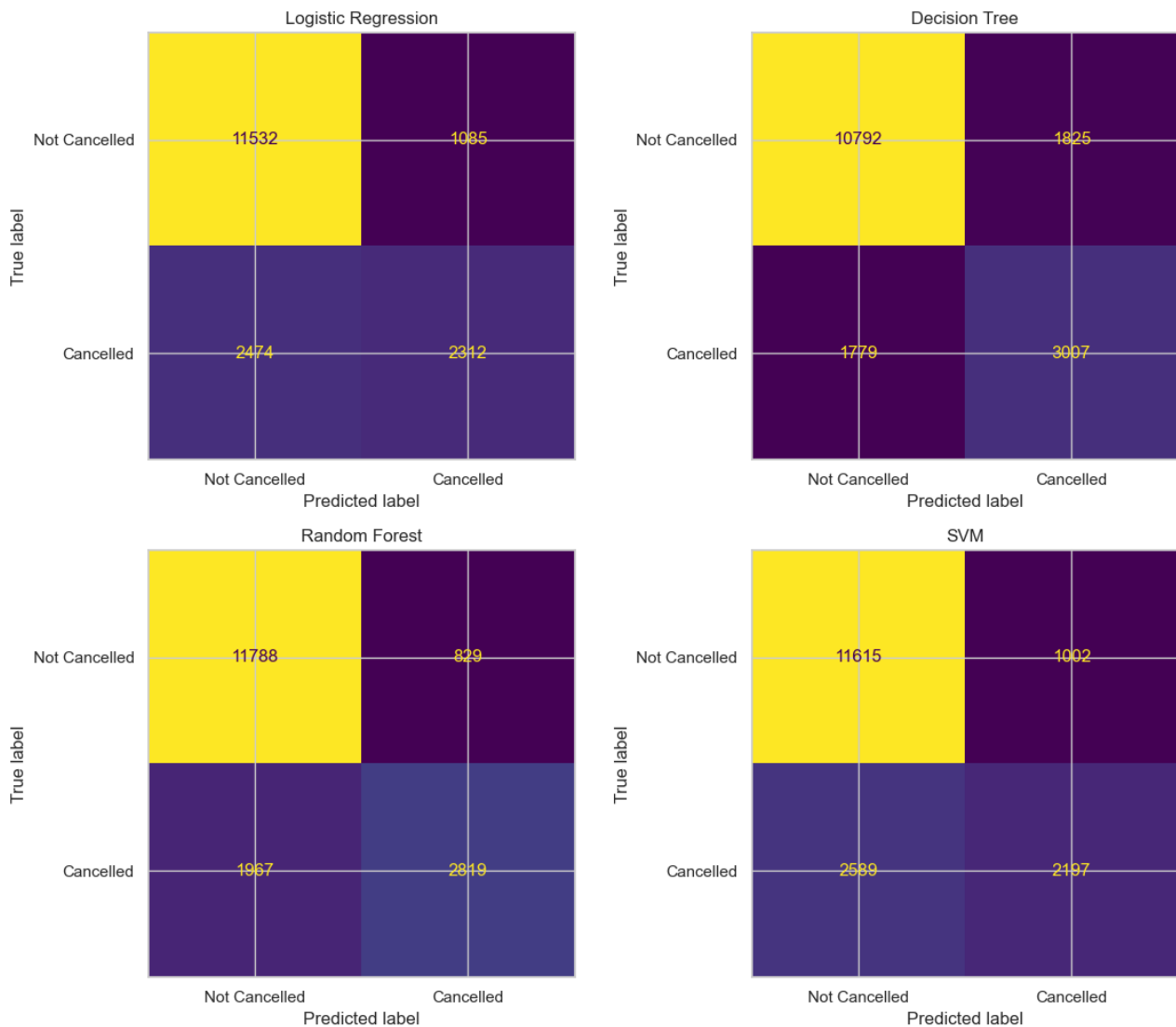


Figure 3. Confusion matrices for the baseline models.

8. Tuned Random Forest Results

The supplied classification report gives the tuned Random Forest class-1 precision as 0.77, recall as 0.60, and F1 score as 0.67. Overall test accuracy is 0.84.

9. Feature Importance

Feature	Importance
lead_time	0.110873
total_stay_cost	0.072963
adr	0.072675
arrival_date_day_of_month	0.058408
total_of_special_requests	0.054437
arrival_date_week_number	0.052803
country_PRT	0.040808
total_nights	0.033432
stays_in_week_nights	0.031792
market_segment_Online TA	0.025404
required_car_parking_spaces	0.025180
arrival_date_year	0.024545
stays_in_weekend_nights	0.022545

Lead time has the largest reported feature importance at 0.110873, followed by total_stay_cost at 0.072963 and adr at 0.072675. These are model-derived importance values and do not establish causation.

10. Streamlit Deployment

The trained models were serialized and integrated into a Streamlit application. Users can select a model, enter booking information, obtain a cancellation prediction, and view an estimated cancellation probability.

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11. Conclusion

The experiment demonstrates an end-to-end machine learning workflow for hotel booking cancellation prediction, from data preparation and exploratory analysis through model comparison, hyperparameter tuning, feature analysis, serialization, and deployment. The tuned Random Forest achieved 84.01% accuracy and a 0.6722 F1 score on the reported test set. The deployed Streamlit application provides an interactive interface for using the trained models.

12. Limitations and Future Work

Future work could investigate decision-threshold tuning, class-imbalance strategies, probability calibration, additional feature engineering, broader hyperparameter optimization, and evaluation on an independent or later-period dataset to assess generalization.